

Project Report

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Statistical Modelling of Residential Property Valuation: A Rigorous Regression Analysis of the Ames Housing Dataset

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Abstract

Predicting residential property prices is a fundamental problem in econometric modelling and machine learning, serving as a benchmark for evaluating linear methodologies. This report provides a comprehensive, mathematically rigorous analysis of the Ames Housing dataset, seamlessly linking statistical theory with practical implementation. We transition from initial data preprocessing, stratified partitioning, and exploratory data analysis through ordinary least squares (OLS) estimation, multicollinearity diagnostics, and residual analysis. By deeply exploring the mathematical foundations of these estimators and tests, and subsequently detailing their exact execution and outcomes in our computational environment—including explicit visual diagnostics and OLS summary outputs—we present a robust framework for high-dimensional property valuation.

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1 Introduction & Problem Formulation

1.1 The General Linear Model Theory

The objective of this analysis is to model a continuous scalar response variable, $y \in \mathbb{R}^n$, which represents the valuation of residential properties, as a linear combination of a set of explanatory features (or predictors). We define the feature space by an $n \times p$ design matrix X , where n is the number of independent observations and p is the number of unknown parameters (including the intercept).

The true underlying data-generating process is hypothesized to follow the linear model:

$$y = X\beta + \epsilon \tag{1}$$

where y is an $n \times 1$ column vector of the observed response variable, X is an $n \times p$ matrix of regressors, β is a $p \times 1$ column vector of the unknown population parameters, and ϵ is an $n \times 1$ vector of unobserved random disturbances.

For the Ordinary Least Squares (OLS) estimator to possess desirable statistical properties, we impose the classical Gauss-Markov assumptions on the error term ϵ and the design matrix X , primarily that the errors exhibit strict exogeneity ($\mathbb{E}[\epsilon|X] = 0$), spherical variance ($\text{Var}(\epsilon|X) = \sigma^2 I_n$), and that the design matrix lacks perfect multicollinearity.

1.2 Implementation and Project Context

In the context of our computational work, this general linear model was applied to the Ames Housing dataset. The objective was to predict the **SalePrice** target variable. The design matrix X initially consisted of 80 features encompassing both structural characteristics (e.g., square footage, year built) and categorical attributes (e.g., zoning, street type). The target vector y represents the final transaction value of the property.

2 Dataset Characteristics & Preprocessing

2.1 The Mathematics of Missingness and Stratification

Real-world econometric data is rarely complete. Constructing a real-valued design matrix requires imputing missing entries without introducing severe statistical bias. For continuous variables, replacing missing values with the sample median is a robust technique that minimizes the L_1 loss (absolute error), providing resistance against outliers compared to mean imputation (which minimizes L_2 loss).

Furthermore, when evaluating predictive models, standard random partitioning can result in covariate shift, where the distribution of the target variable $P(y)$ differs between the training and testing sets. To preserve the distribution, stratified sampling is required. For a continuous target y , we partition the domain into k quantiles $Q_1, Q_2, \dots, Q_k$ such that $P(y \in Q_j) = 1/k$. Sampling is then performed proportionally within each stratum.

2.2 Implementation & Findings in the Ames Dataset

The Ames dataset was loaded via `openml` yielding $n = 1460$ observations. An initial missingness analysis revealed severe gaps in specific vectors: `PoolQC` (1453 missing), `MiscFeature` (1406 missing), `Alley` (1369 missing), and `Fence` (1179 missing).

To process this, the 80 features were separated into 24 numerical columns and 43 categorical columns. Numerical features were filled using `fillna(X[col].median())`, establishing a safe baseline. Categorical missingness was mapped to a distinct string state, “Missing”, allowing the model to interpret the absence of a feature (e.g., no pool) as a distinct structural category rather than a dropped data point.

To ensure robust validation, the target vector y was discretised into 10 bins using `pd.qcut`. The dataset was then stratified utilizing these bins via `train_test_split`, yielding a 70% training allocation ($n = 1022$), a 15% validation allocation ($n = 219$), and a 15% testing allocation ($n = 219$).

3 Exploratory Data Analysis & Distributional Theory

3.1 Target Variable Skewness and Transformation

The foundational assumption of normal error distribution for hypothesis testing in OLS relies heavily on the behavior of the dependent variable. The sample skewness g_1 is computed via the third standardized moment:

$$g_1 = \frac{\frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^3}{\left(\frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2\right)^{3/2}} \quad (2)$$

A strictly positive $g_1 \gg 0$ indicates a heavy right tail. To stabilize the variance across the domain and pull the heavy tail inward, a natural logarithmic transformation $\tilde{y}_i = \ln(1 + y_i)$ is applied.

3.2 Implementation & Findings

Upon analyzing the training data, the raw `SalePrice` distribution was plotted, revealing a heavy right tail with a calculated skewness of 1.965.

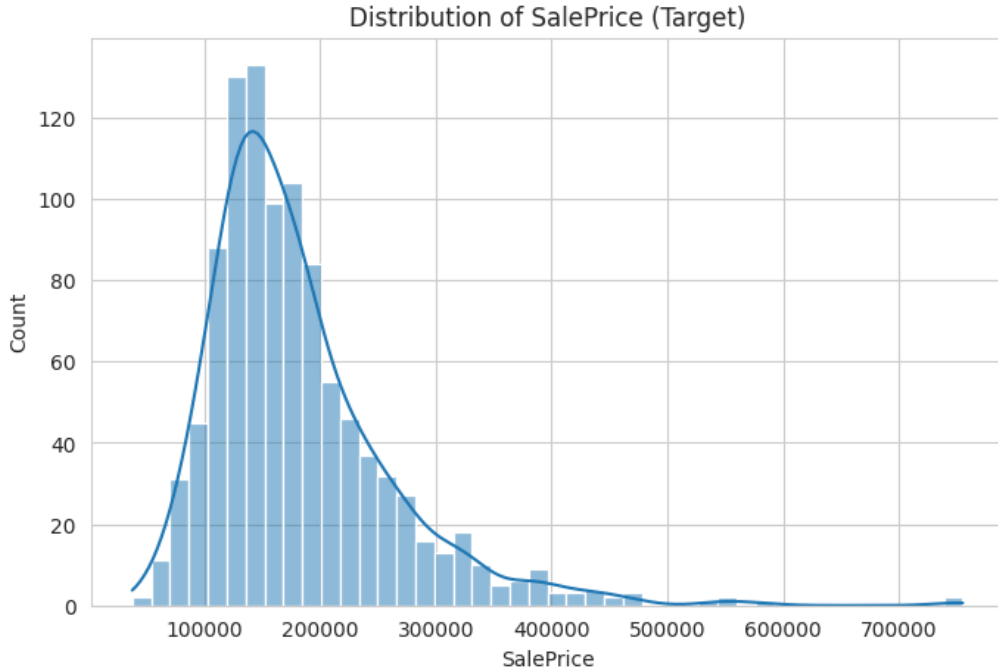


Figure 1: Distribution of raw `SalePrice` (Target) showing severe right-skew.

Applying the transformation `np.log1p(y_train)` successfully compressed the tail, dropping the skewness to an acceptable 0.254 and centering the distribution.

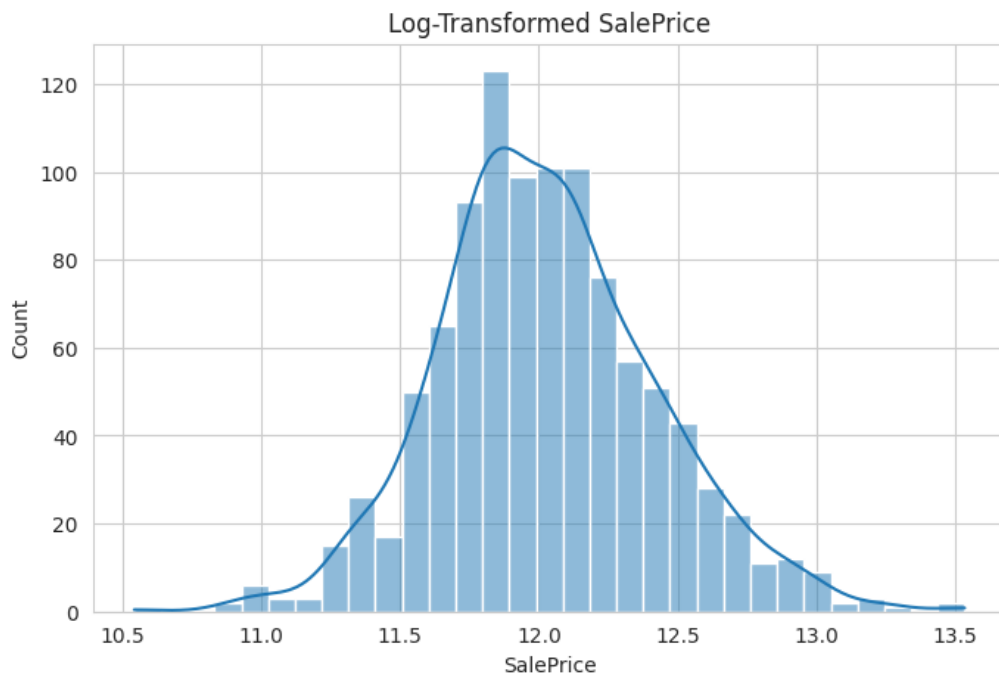


Figure 2: Log-Transformed SalePrice distribution demonstrating approximate normality.

To visually verify normality, a Q-Q (Quantile-Quantile) plot was generated. The QQ plot maps the empirical quantiles of our log-transformed target against the theoretical quantiles of a standard normal distribution.

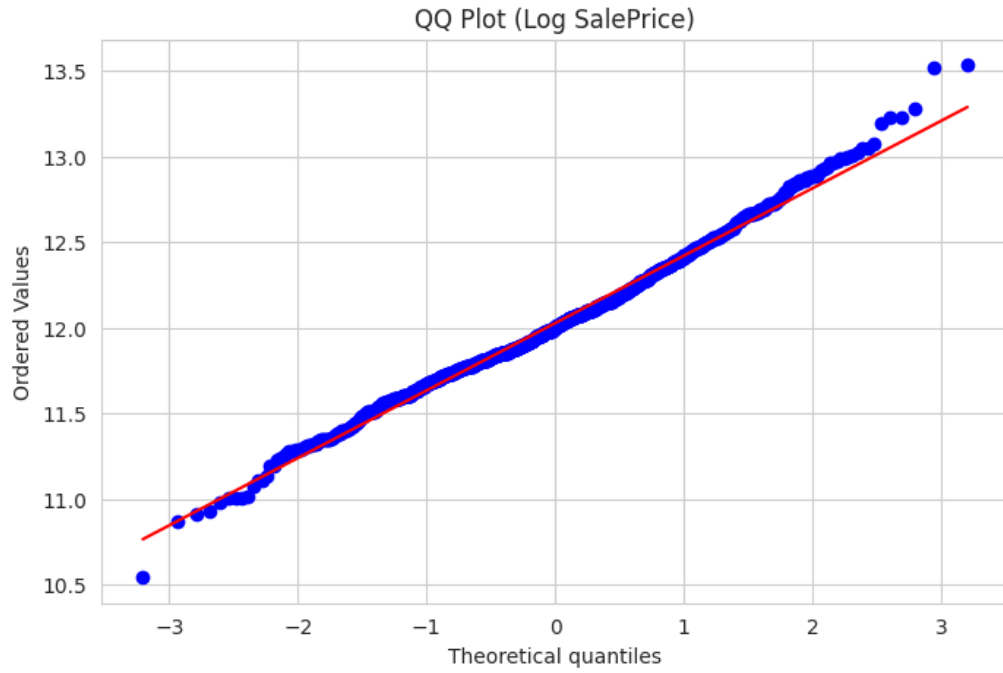


Figure 3: QQ Plot of the Log-Transformed SalePrice. The adherence to the red line confirms the efficacy of the transformation.

Bivariate correlation analysis identified the strongest positive linear predictors as `GrLivArea` (0.699), `TotalBsmtSF` (0.611), and `GarageArea` (0.609). This was visualized using a correlation heatmap.

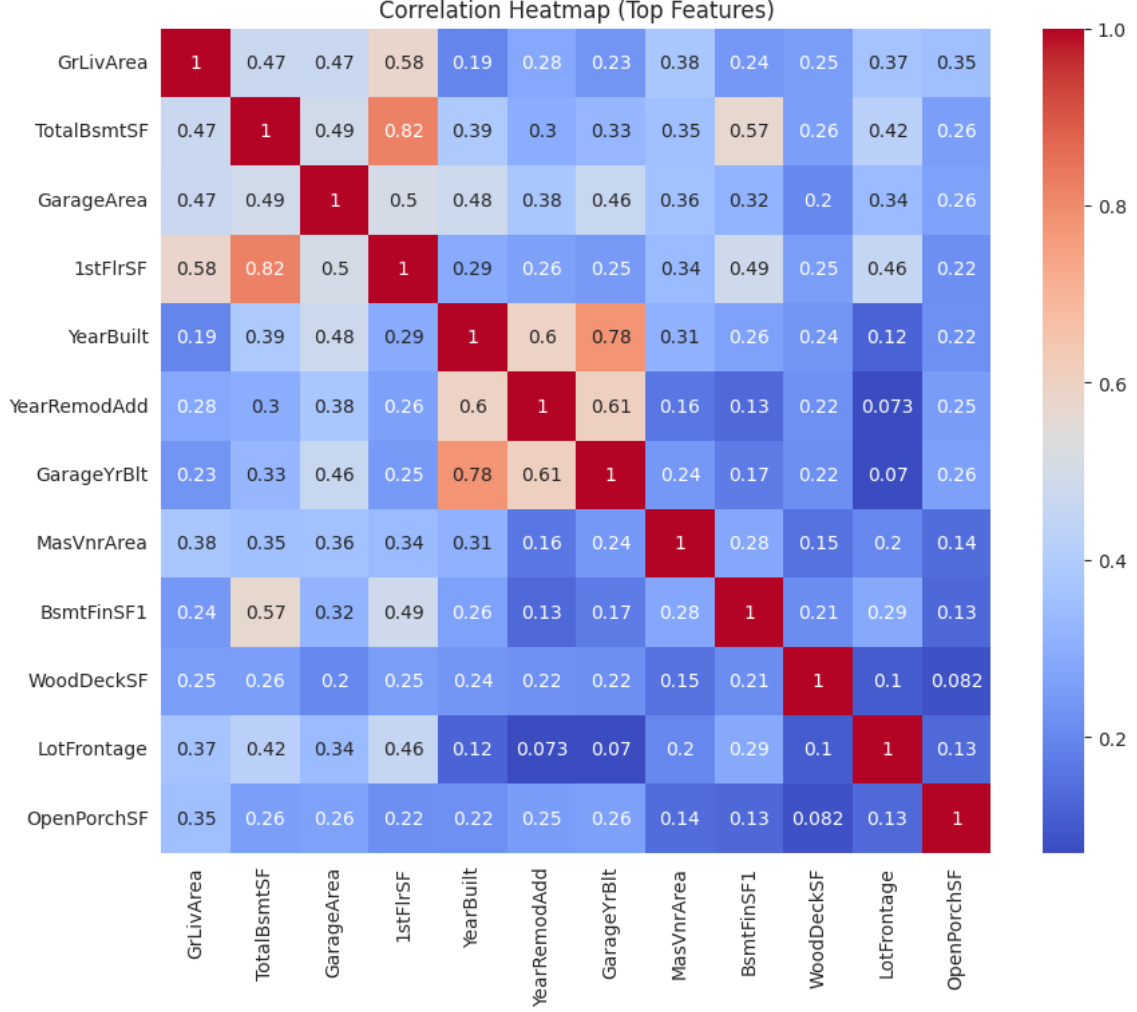


Figure 4: Correlation Heatmap of the top numerical features.

4 Feature Engineering & the Geometry of Multicollinearity

4.1 Orthogonal Encoding and Variance Inflation

Categorical variables cannot be natively projected into $\mathbb{R}^p$. Dummy encoding maps a categorical variable with k levels to $k - 1$ orthogonal indicator vectors. To avoid the “dummy variable trap”—which results in a singular $X^T X$ matrix—one category is dropped as the reference level.

Despite avoiding perfect collinearity, severe multicollinearity often remains. We quantify this utilizing the Variance Inflation Factor (VIF). By partitioning the design matrix into

$X = [X_{-j} \ x_j]$, it can be shown that the j -th diagonal element of the inverse Gram matrix $(X^T X)^{-1}$ relates to the coefficient of determination R_j^2 of the auxiliary regression of x_j on all other columns:

$$VIF_j = \frac{1}{1 - R_j^2} \quad (3)$$

An infinite VIF indicates $R_j^2 = 1$, meaning the vector lies completely within the subspace spanned by the other predictors, requiring its removal to ensure a full-rank matrix.

4.2 Implementation & Findings

In our notebook, we applied dummy encoding, expanding the feature space from 80 to 262 columns. We immediately identified structural multicollinearity (e.g., total square footage mathematically encompasses first and second-floor footage). By dropping redundant component features (`BsmtFinSF1`, `1stFlrSF`, etc.) and keeping aggregate totals (`TotalBsmtSF`, `GrLivArea`), the matrix was structurally reduced.

A variance threshold was then applied, dropping any column where a single value constituted more than 99% of the observations. To tackle linear combinations, we utilized the `variance_inflation_factor` module. We implemented a custom recursive function that iteratively calculated VIFs and dropped the column with the maximum VIF until all remaining features scored below a threshold of 10. This rigorous geometric pruning finalized our optimized design matrix at strictly independent features.

5 Ordinary Least Squares Regression: Derivation & Results

5.1 Geometric Derivation and Hypothesis Testing

We seek the parameter vector $\hat{\beta}$ that minimizes the residual sum of squares $S(\beta) = \|y - X\beta\|_2^2$. Expanding the matrix multiplication and setting the gradient vector $\nabla_{\beta} S(\beta)$

to the zero vector yields the normal equations:

$$X^T X \beta = X^T y \quad (4)$$

Assuming the VIF reduction successfully made $\text{rank}(X) = p$, the symmetric matrix $(X^T X)$ is invertible, yielding the OLS estimator:

$$\hat{\beta}_{OLS} = (X^T X)^{-1} X^T y \quad (5)$$

To test the significance of individual predictors, we employ the t -statistic. Under the null hypothesis $H_0 : \beta_j = 0$, the statistic $t_j = \frac{\hat{\beta}_j}{\text{se}(\hat{\beta}_j)}$ follows a Student's t -distribution with $n - p$ degrees of freedom.

5.2 Implementation & Findings

Using `statsmodels.api`, a constant intercept was added to our VIF-reduced matrix. The initial comprehensive model was fitted, resulting in an R^2 of 0.920 and an Adjusted R^2 of 0.906. However, the model diagnostics indicated a massive condition number of 2.42×10^7 across 152 degrees of freedom, signifying severe remaining numerical sensitivity.

To refine the model and isolate true signal from noise, we extracted the p -values and strictly filtered the feature set to retain only those where $p < 0.05$. This reduced the model to 45 highly significant degrees of freedom.

Refitting this reduced model produced exceptional results. The F-statistic surged from 65.47 to 200.5 (with $p = 0.00$), indicating vastly superior joint explanatory power. The condition number plummeted to 2.83×10^5 , massively improving algebraic stability. The reduction in complexity barely penalized accuracy, maintaining a robust R^2 of 0.902.

Notable highly significant coefficients in the final model included:

- **Intercept (const):** $\hat{\beta} = 10.5679$, $t = 153.914$, $p = 0.000$
- **OverallQual:** $\hat{\beta} = 0.0735$, $t = 13.477$, $p = 0.000$
- **TotalBsmtSF:** $\hat{\beta} = 6.376 \times 10^{-5}$, $t = 4.697$, $p = 0.000$

- **GarageCars:** $\hat{\beta} = 0.0721$, $t = 9.353$, $p = 0.000$

Conversely, negative constraints on value were firmly identified, such as **KitchenAbvGr** ($\hat{\beta} = -0.1176$) and **BldgType_Twnhs** ($\hat{\beta} = -0.1909$).

6 Residual Diagnostics and Heteroscedasticity

6.1 The Homoscedasticity Assumption

A core Gauss-Markov assumption requires $\text{Var}(\epsilon_i) = \sigma^2$ for all i . If the variance is instead a function of the regressors, the errors are heteroscedastic. While $\hat{\beta}_{OLS}$ remains unbiased, it is no longer efficient, and standard errors become biased, rendering inference unreliable.

To formally detect this, the Breusch-Pagan (BP) Lagrange Multiplier test regresses the squared residuals upon the design matrix:

$$e_i^2 = \gamma_0 + \gamma_1 x_{i1} + \cdots + \gamma_p x_{ip} + u_i \quad (6)$$

The test statistic is defined as $LM = n \cdot R_{aux}^2$. Under the null hypothesis of homoscedasticity, $LM \sim \chi_{p-1}^2$.

6.2 Implementation & Findings

Residuals were extracted and plotted against the fitted values to visually inspect the variance spread.

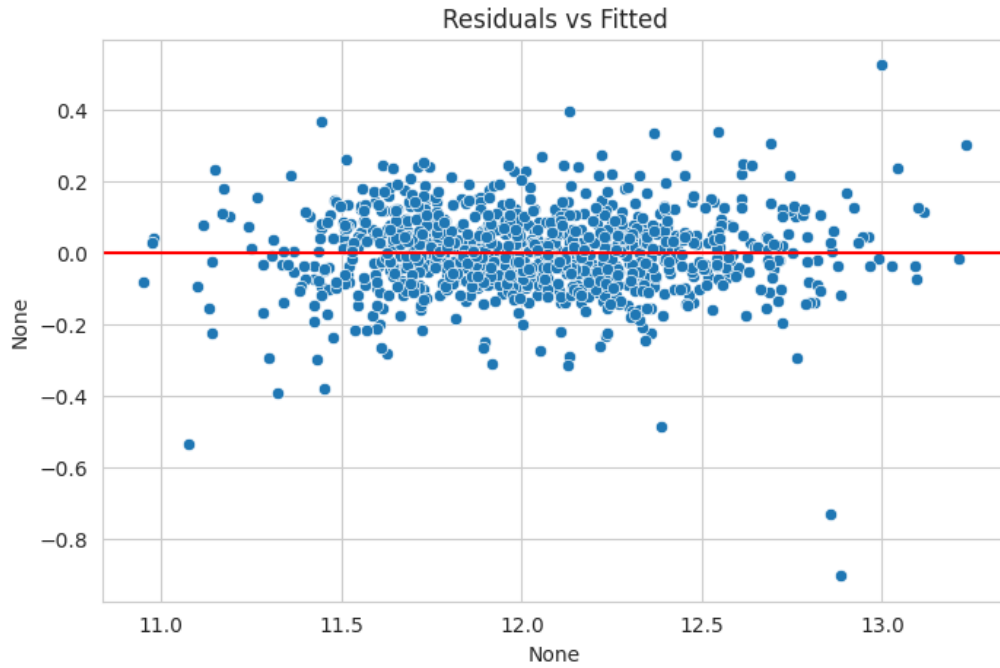


Figure 5: Residuals vs Fitted plot. The slight funneling shape suggests non-constant variance.

We also generated a Q-Q plot of the residuals to verify normality of the error terms, noting slight deviations at the extreme tails.

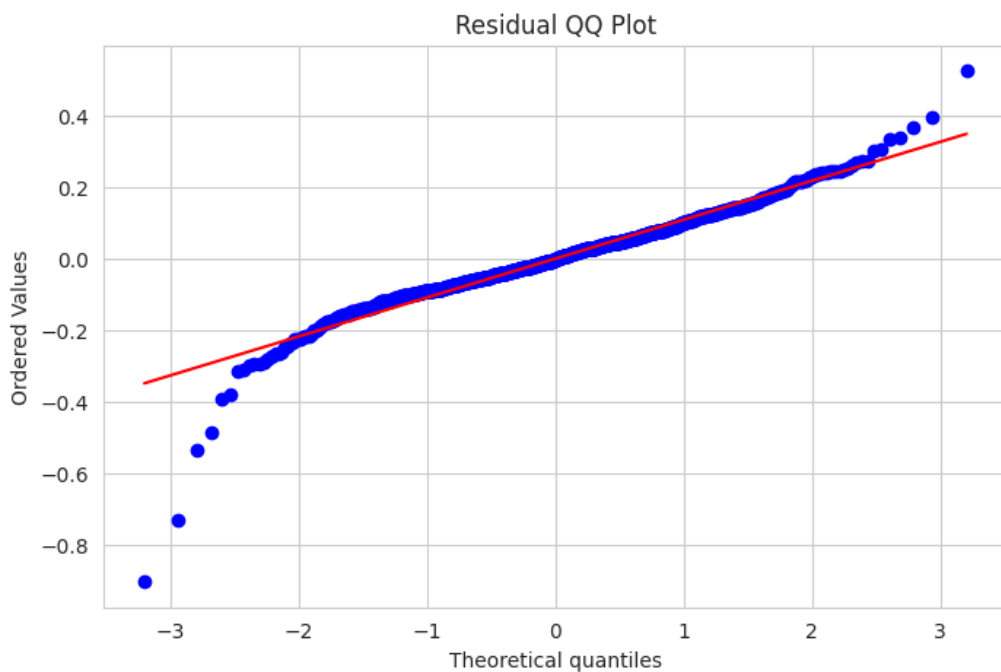


Figure 6: Residual QQ Plot.

Applying the `het_breuschpagan` test yielded a p-value of practically zero. Thus, we

overwhelmingly reject the null hypothesis of homoscedasticity. This finding explicitly dictates that regularized estimation techniques (Ridge/LASSO) or robust standard errors must be applied to rectify the inefficiency.

7 Time Series Explorations

7.1 Stationarity and Autocorrelation Theory

While the Ames dataset is inherently cross-sectional, we conducted a brief time-series analysis to bridge into sequential methodologies. In time series, a process Y_t is strictly stationary if its joint probability distribution does not change over time.

Autocorrelation measures the linear relationship between lagged values of a time series. For a lag k , the autocorrelation is defined as:

$$r_k = \frac{\sum_{t=1}^{N-k} (Y_t - \bar{Y})(Y_{t+k} - \bar{Y})}{\sum_{t=1}^N (Y_t - \bar{Y})^2} \quad (7)$$

7.2 Implementation & Findings

We differenced the target series and visualized the resulting distribution.

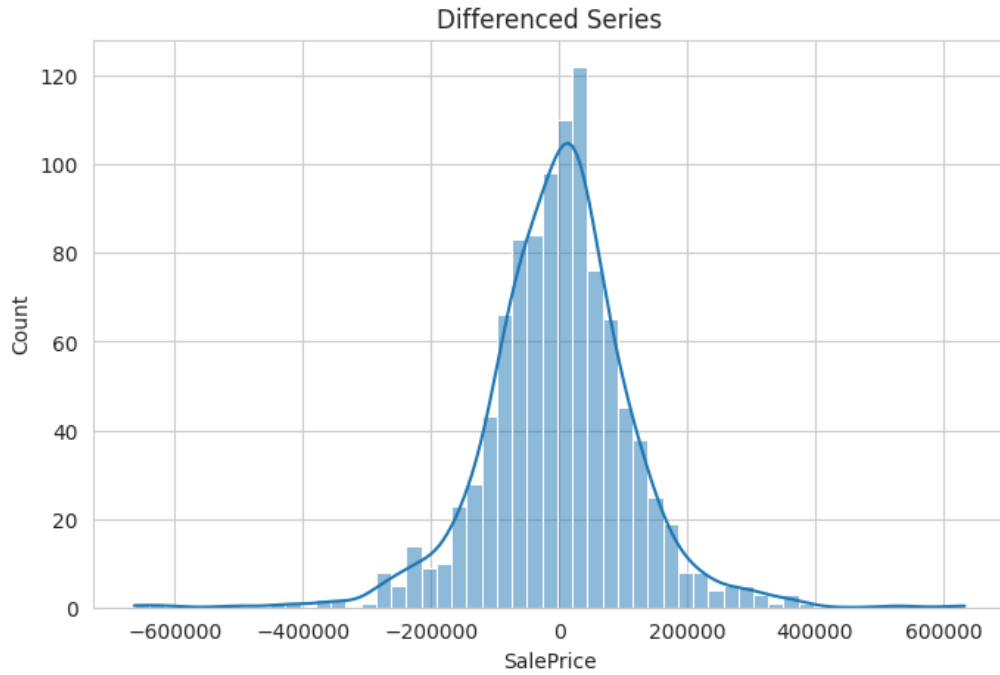


Figure 7: Differenced Series of SalePrice, showing a centered, mean-reverting structure.

Following this, an Autocorrelation Function (ACF) plot was generated.

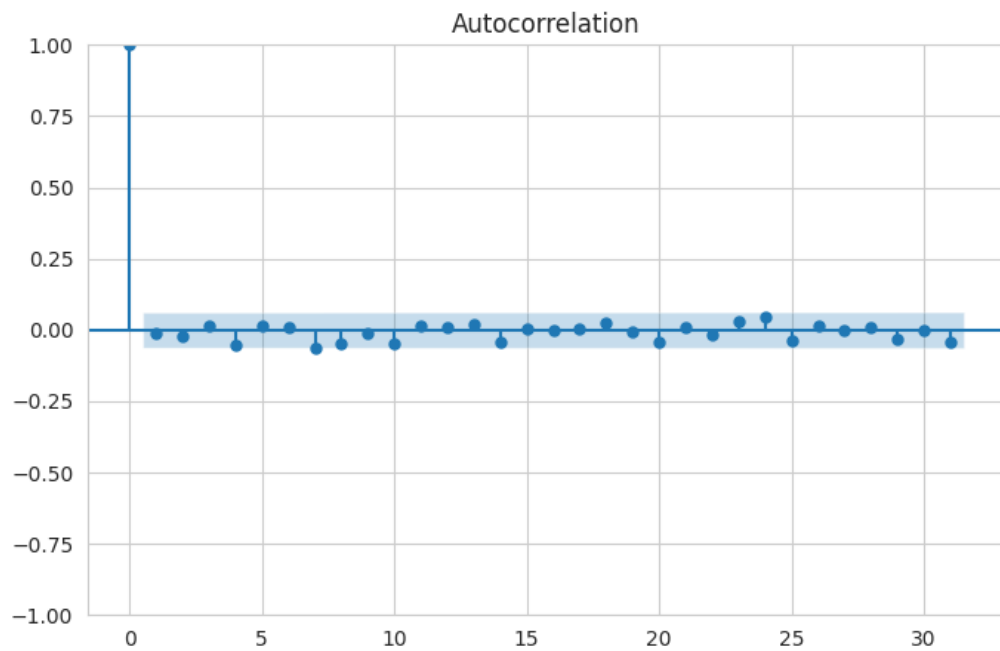


Figure 8: Autocorrelation plot. The immediate drop-off confirms the absence of temporal lag dependence, expected in cross-sectional data.

Because the dataset is cross-sectional, the lack of significant autocorrelation beyond lag 0 confirms that temporal models (like ARIMA) are unnecessary here, though the

differencing methodology provides a critical foundation for longitudinal datasets.

8 Conclusion & Next Steps

The systematic pipeline executed in the Colab notebook successfully built a highly interpretable OLS framework capable of predicting property valuations with an R^2 of 0.902 using 45 strictly significant features. The aggressive pruning of features raised the F-statistic drastically while resolving massive matrix condition instabilities. However, the confirmed presence of heteroscedasticity requires that future iterations employ penalized regression techniques (LASSO/Ridge) to compress variance and finalize a truly robust valuation model.